

15 Surfaces and Surface Integrals

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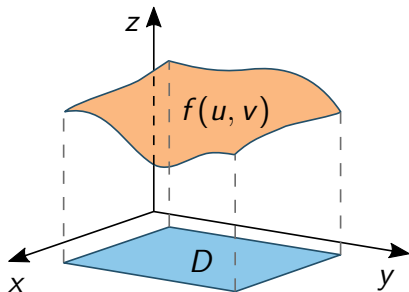
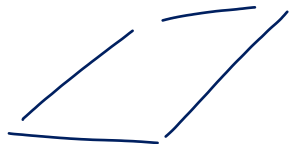
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MATH2111 Several Variable Calculus, T1 2025

Intuition

The graph of $z = f(x, y)$ defined on D has a parametrisation

$$\underline{\mathbf{r}(u, v) = u\mathbf{i} + v\mathbf{j} + f(u, v)\mathbf{k}}, \quad (u, v) \in D.$$



Example 15.1

Find a parametrisation of the plane

$$\underline{3x + 2y + z = 1} \quad z = 1 - 3x - 2y$$

$$\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + (1 - 3x - 2y)\mathbf{k} = (x, y, (1 - 3x - 2y)),$$

$x, y \in \mathbb{R}$

Example 15.2

Find a parametrisation of the surface

$$z^2 = x^2 + y^2, \quad z \geq 0.$$

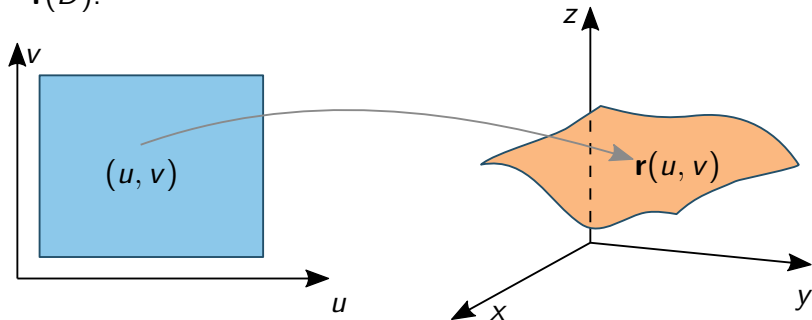
$$\mathbf{r}(x, y) = (x, y, \sqrt{x^2 + y^2})$$

Parametrisation of a surface

A parametrisation of a surface $\mathcal{S}$ is a function $\mathbf{r} : D \rightarrow \mathbb{R}^3$, where $D \subseteq \mathbb{R}^2$,

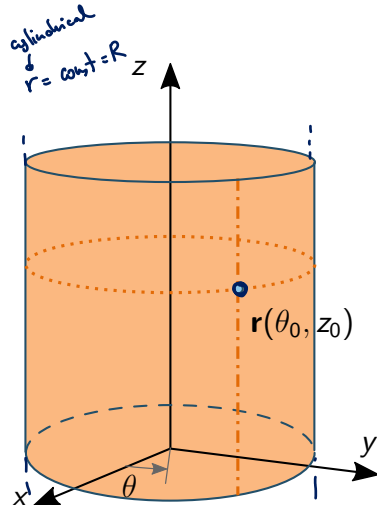
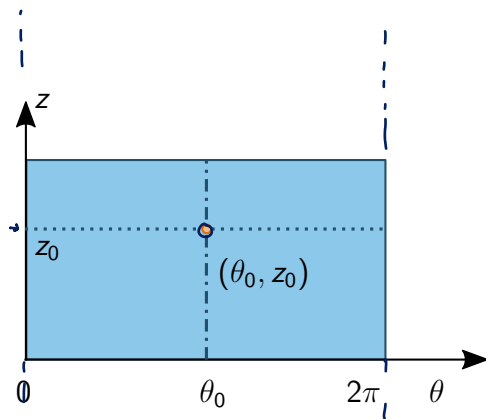
$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k},$$

such that $\mathcal{S} = \mathbf{r}(D)$.



We normally assume that the parametrisation of the surface is one-to-one on the interior of D .

Parametrisation of a cylinder



The cylinder of a given radius R centered at the z -axis can be parametrised as $\mathbf{r}(\theta, z) = (R \cos \theta, R \sin \theta, z)$, $0 \leq \theta \leq 2\pi$.

Parametrisation of a cone

Consider the cone $z^2 = x^2 + y^2$, $z \geq 0$.

From the equation $z = \sqrt{x^2 + y^2}$ we have

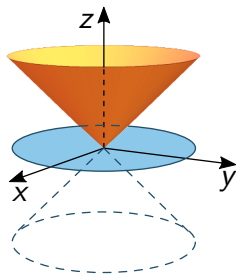
$$\mathbf{r}(u, v) = u\mathbf{i} + v\mathbf{j} + \sqrt{u^2 + v^2}\mathbf{k}.$$

Using cylindricals:

$$\mathbf{r}(\theta, r) = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j} + r \mathbf{k}, \quad \theta \in [0, 2\pi], r \geq 0. \quad (1)$$

In sphericals we have $\phi = \frac{\pi}{4}$:

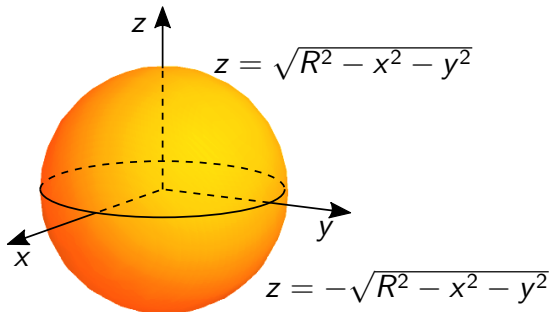
$$\mathbf{r}(\theta, \rho) = \underbrace{\frac{1}{\sqrt{2}}\rho \cos \theta}_{\text{}} \mathbf{i} + \underbrace{\frac{1}{\sqrt{2}}\rho \sin \theta}_{\text{}} \mathbf{j} + \underbrace{\frac{1}{\sqrt{2}}\rho}_{\text{}} \mathbf{k}, \quad \theta \in [0, 2\pi], \rho \geq 0.$$



Parametrisation of a Sphere

From the Cartesian coordinate equations the sphere is defined by two functions:

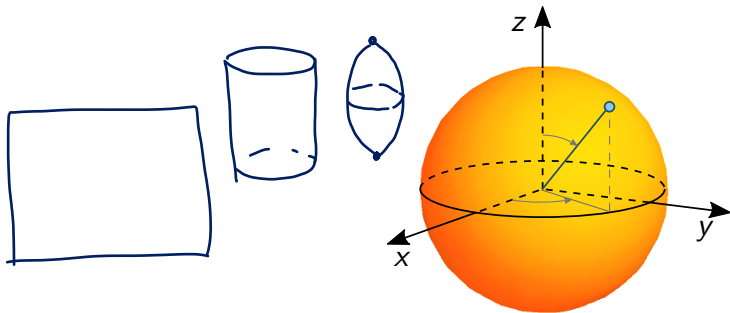
$$z = \pm\sqrt{R^2 - x^2 - y^2}.$$



Hence we have two different parametrisations for each patch:

$$\mathbf{r}_+(u, v) = u\mathbf{i} + v\mathbf{j} + \sqrt{R^2 - u^2 - v^2}\mathbf{k},$$

$$\mathbf{r}_-(u, v) = u\mathbf{i} + v\mathbf{j} - \sqrt{R^2 - u^2 - v^2}\mathbf{k}.$$



In sphericals the equation is $\rho = R$, so we have the parametrisation

$$\mathbf{r}(\theta, \phi) = \underline{R \cos \theta \sin \phi \mathbf{i} + R \sin \theta \sin \phi \mathbf{j} + R \cos \phi \mathbf{k}}.$$

Note that for $\phi = 0$ we have

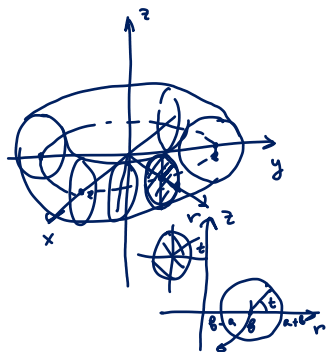
$$\mathbf{r}(\theta, 0) = \underline{R \mathbf{k}}.$$

The mapping is one-to-one on the interior of D , but not on the boundary.

Parametrisation of a torus

Example 15.3

Find a parametrisation of the surface of a torus with minor radius a and major radius $b > a > 0$ that is (rotationally) symmetric about the z -axis and is also symmetric with respect to the xy -plane.



$$\begin{cases} z^2 + (r - b)^2 = a^2 \\ z = \pm \sqrt{a^2 - (r - b)^2} \end{cases}$$

$$z = a \sin t$$

$$r = a \cos t + b$$

$\Downarrow$

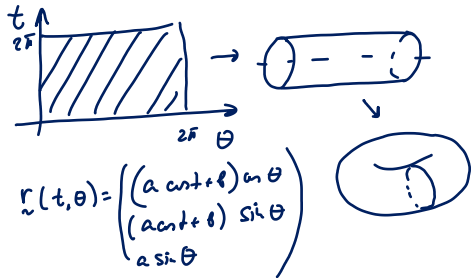
$$z = a \sin t$$

$$x = (a \cos t + b) \cos \theta$$

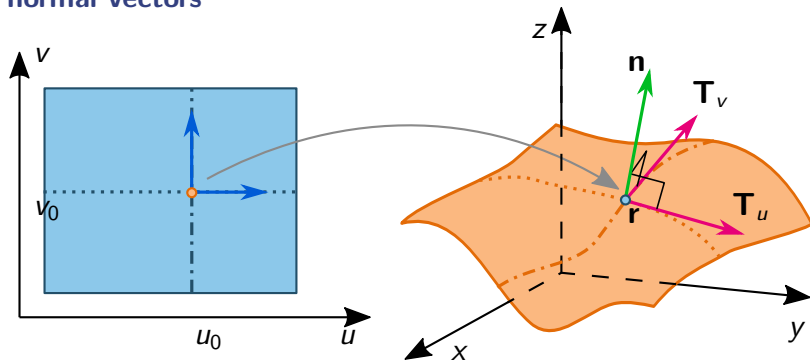
$$y = (a \cos t + b) \sin \theta$$

$$\theta \in [0, 2\pi]$$

$$t \in [0, 2\pi]$$



Tangent and normal vectors



If a surface is parametrised by a C^1 function $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$, we can fix $v = v_0$ and consider the path $\tilde{\mathbf{c}}(u) = \mathbf{r}(u, v_0)$. We can obtain the tangent vector to this path as

$$\mathbf{T}_u = \frac{\partial \mathbf{r}}{\partial u} = \left(\frac{\partial x(u, v_0)}{\partial u}, \frac{\partial y(u, v_0)}{\partial u}, \frac{\partial z(u, v_0)}{\partial u} \right).$$

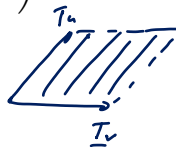
Tangent and normal vectors

Similarly, fixing $u = u_0$ we obtain the tangent to the path that corresponds to the grid line $u = u_0$ in D and its image:

$$\mathbf{T}_v = \frac{\partial \mathbf{r}}{\partial v} = \left(\frac{\partial x(u_0, v)}{\partial v}, \frac{\partial y(u_0, v)}{\partial v}, \frac{\partial z(u_0, v)}{\partial v} \right).$$

We can compute a normal vector,

$$\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v.$$



Note that the length of this vector is the area of the parallelogram with sides $\mathbf{T}_u$ and $\mathbf{T}_v$.

Example 15.4

What are the tangent and normal vectors to the surface defined as the graph of some function $z = f(x, y)$?

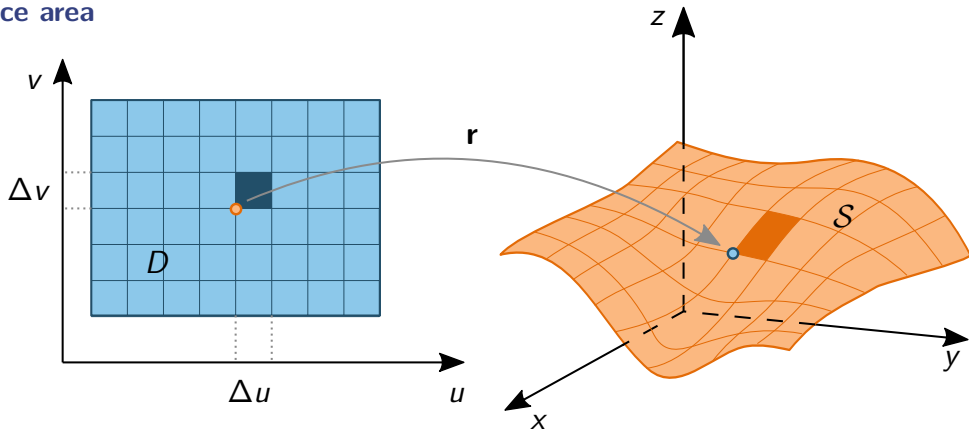
$$\vec{r}(x, y) = \begin{pmatrix} x \\ y \\ f(x, y) \end{pmatrix}$$

$$\vec{T}_x = \frac{\partial \vec{r}}{\partial x} = \begin{pmatrix} 1 \\ 0 \\ \frac{\partial f(x, y)}{\partial x} \end{pmatrix}$$

$$\vec{T}_y = \frac{\partial \vec{r}}{\partial y} = \begin{pmatrix} 0 \\ 1 \\ \frac{\partial f(x, y)}{\partial y} \end{pmatrix}$$

$$\vec{n} = \vec{T}_x \times \vec{T}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & \frac{\partial f}{\partial x} \\ 0 & 1 & \frac{\partial f}{\partial y} \end{vmatrix} = \begin{pmatrix} -\frac{\partial f}{\partial x} \\ -\frac{\partial f}{\partial y} \\ 1 \end{pmatrix}$$

Surface area



Let $\mathbf{r}$ be a C^1 parametrisation of a surface S with a simple parameter domain D . Then the area of this surface is

$$\text{Area}(S) = \iint_S dS = \iint_D \underbrace{\|\mathbf{T}_u \times \mathbf{T}_v\|}_{dS} du dv.$$

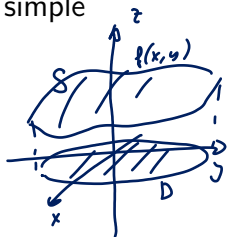
Example 15.5 (Area of the graph of a function)

Evaluate the area of the graph of some function $f(x, y)$ defined on a simple two-dimensional region D .

$$\text{Area}(S) = \iint_D \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1} \, dx \, dy$$

From Example 15.4 $\boxed{\underline{T}_u \times \underline{T}_v} = \begin{pmatrix} -\frac{\partial f}{\partial x} \\ -\frac{\partial f}{\partial y} \\ 1 \end{pmatrix}$

$$\Rightarrow \|\underline{T}_u \times \underline{T}_v\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1}$$



$$\text{Area}(\mathcal{S}) = \iint_D \sqrt{\left(\frac{\partial f(u, v)}{\partial u}\right)^2 + \left(\frac{\partial f(u, v)}{\partial v}\right)^2 + 1} du dv.$$

Example 15.6

Find the area of a sphere of radius R .

By symmetry, the area of a sphere = $2 \text{ area}(S)$,
where S is the upper hemisphere

$$\mathbf{r}(x,y) = \begin{pmatrix} x \\ y \\ \sqrt{R^2 - x^2 - y^2} \end{pmatrix} \quad f(x,y) = \sqrt{R^2 - x^2 - y^2}$$

$$2 \text{ area}(S) = 2 \iint_D \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1} \, dx \, dy \quad \textcircled{=}$$

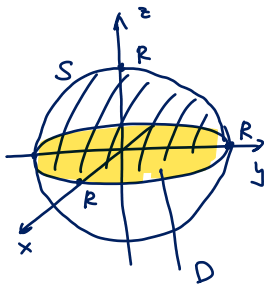
$$\sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1} = \sqrt{\frac{x^2}{R^2 - x^2 - y^2} + \frac{y^2}{R^2 - x^2 - y^2} + 1} = \sqrt{\frac{x^2 + y^2}{R^2 - x^2 - y^2} + 1} = \sqrt{\frac{x^2 + y^2 + R^2 - x^2 - y^2}{R^2 - x^2 - y^2}} = \frac{R}{\sqrt{R^2 - x^2 - y^2}}$$

$$\textcircled{=} 2 \iint_D \frac{R}{\sqrt{R^2 - x^2 - y^2}} \, dx \, dy = 2 \int_0^{2\pi} d\theta \int_0^R \frac{R}{\sqrt{R^2 - r^2}} \cdot r \, dr = 2 \cdot 2\pi \cdot R \cdot \left(-\sqrt{R^2 - r^2} \right) \Big|_{r=0}^{r=R}$$

$$= 4\pi \cdot R \cdot \sqrt{R^2}$$

$$= 4\pi R^2$$

$$\left| \sqrt{R^2 - r^2} \right|' = -\left(\frac{r}{\sqrt{R^2 - r^2}} \right)$$



Scalar Surface Integral

Let $\mathcal{S} = \mathbf{r}(D)$ be a surface parametrised by a C^1 function $\mathbf{r}$ defined on a simple two-dimensional region D . For a scalar function f defined on $\mathcal{S}$ the surface integral of f over $\mathcal{S}$ is the double integral

$$\iint_{\mathcal{S}} f \, dS = \iint_D f(\mathbf{r}(u, v)) \underbrace{\|\mathbf{T}_u \times \mathbf{T}_v\|}_{\text{like the Jacobian, } \|\mathbf{r}'(t)\|} \, du \, dv.$$

Example 15.7

Evaluate the integral

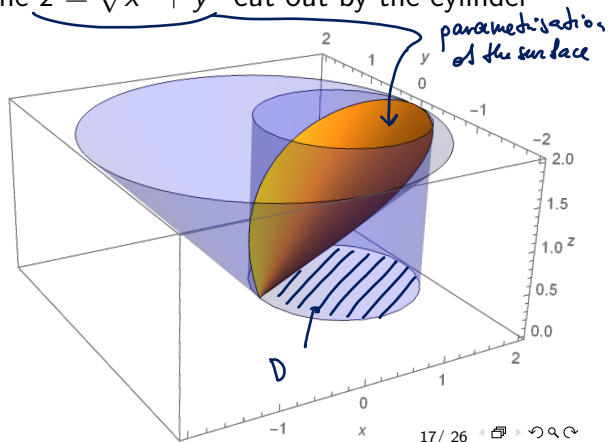
$$I = \iint_S \underbrace{(y^2 z^2 + z^2 x^2 + x^2 y^2)}_{f(x,y,z)} dS,$$

where S is the part of the surface of the cone $z = \sqrt{x^2 + y^2}$ cut out by the cylinder $x^2 + y^2 - 2ax = 0$. $\leftarrow$ region D

$$I = \iint_D \underbrace{(y^2(x^2+y^2) + (x^2+y^2)x^2 + x^2y^2)}_{f(r)} \sqrt{2} \, dx \, dy$$

$$\underline{r}(x,y) = \begin{pmatrix} x \\ y \\ \sqrt{x^2+y^2} \end{pmatrix} = \underline{s}(x,y)$$

$$\begin{aligned} \|\underline{T}_x \times \underline{T}_y\| &= \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} = \sqrt{\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} + 1} \\ &= \sqrt{\frac{x^2+y^2}{x^2+y^2} + 1} = \sqrt{2} \end{aligned}$$



$$\begin{aligned}
 I &= \sqrt{2} \iint_D ((x^2+y^2)^2 + x^2y^2) dx dy \\
 &= \sqrt{2} \int_{-\pi/2}^{\pi/2} d\theta \int_0^{2a\cos\theta} (r^4 + r^4 \cos^2\theta \sin^2\theta) r dr \\
 &= \dots = \frac{29\sqrt{2}}{8} \pi a^6
 \end{aligned}$$

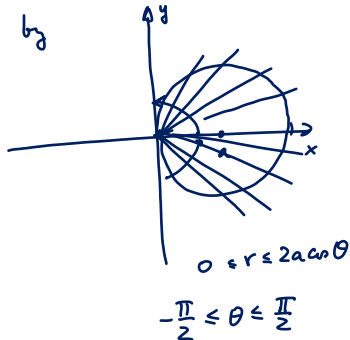
D is the region bounded by

$$x^2 + y^2 - 2ax = 0$$

$$(x-a)^2 + y^2 = a^2$$

$$r^2 - 2ar\cos\theta = 0$$

$$\Rightarrow r = 2a\cos\theta$$



Similarity between the scalar path and surface integrals

- The scalar path and surface integrals are conceptually similar, and can both be interpreted as the evaluation of the total mass of the object from a known density function.
- The value of the scalar surface integral $\iint_S f \, dS$ does not depend on the parametrisation of the surface S and hence on the choice of the *orientation* of the normals.
- If a surface S consists of a finite number of non-overlapping surfaces $S_1, S_2, \dots, S_n$, then the surface integral of a scalar function f over this surface can be evaluated as the sum

$$\iint_S f \, dS = \sum_{i=1}^n \iint_{S_i} f \, dS$$

Vector Surface Integral

For a vector field $\mathbf{F}$ defined on a parametrised C^1 surface $S = \mathbf{r}(D)$ the vector surface integral of $\mathbf{F}$ over S is the following double integral,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS \\ &= \iint_{D)} \underbrace{\mathbf{F} \cdot \hat{\mathbf{n}}}_{\text{dot product}} \|\mathbf{T}_u \times \mathbf{T}_v\| du dv.\end{aligned}$$



surface should be orientable!

Here $\hat{\mathbf{n}}$ is the unit normal vector that can be explicitly calculated as

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{\|\mathbf{n}\|} = \frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} \text{ or } \hat{\mathbf{n}} = -\frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} = \frac{\mathbf{T}_v \times \mathbf{T}_u}{\|\mathbf{T}_u \times \mathbf{T}_v\|},$$

unit normal

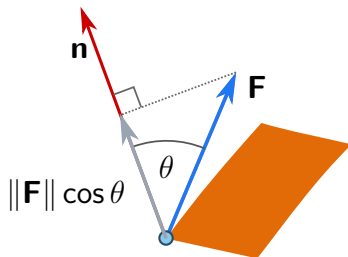
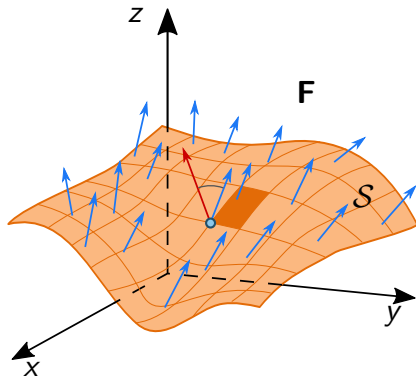
$$\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)}{\|\mathbf{T}_u \times \mathbf{T}_v\|}$$

sign depends on the orientation of normals!

depending on the orientation of the normal.

$$\iint_D \mathbf{F} \cdot \hat{\mathbf{n}} \|\mathbf{T}_u \times \mathbf{T}_v\| du dv = \pm \iint_D \frac{\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)}{\|\mathbf{T}_u \times \mathbf{T}_v\|} \cdot \|\mathbf{T}_u \times \mathbf{T}_v\| du dv = \pm \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) du dv$$

Geometric Interpretation



The vector surface integral can be understood as the *flux* of a vector field across the surface. In other words, if the vector field models the flow of a liquid, then the flux is the amount of liquid per unit time that flows through the surface.

Example 15.8

Evaluate $\iint_{\mathcal{S}} \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$ where $\mathcal{S}$ is the surface of the unit sphere $x^2 + y^2 + z^2 = 1$ and $\hat{\mathbf{n}}$ is directed outwards for $\mathbf{F} = (x, y, z)$.

Example 15.9

Let $\mathcal{S}$ be the surface of the cone $z^2 = x^2 + y^2$, bounded by $0 \leq z \leq h$, with normals pointing outwards. Let $\mathbf{F} = (y - z, z - x, x - y)$. Evaluate

$$\iint_{\mathcal{S}} \mathbf{F} \cdot d\mathbf{S}.$$

Vector line vs surface integrals

- In contrast to the scalar line and surface integrals, the vector line integral and the vector surface integral are conceptually different. We 'project' the vector field on the tangent direction in the case of the path integral, but for the surface integral the 'projection' is onto the normal vector.
- Reversing the orientation of the normals changes the sign of the vector surface integral. However, given that the orientation is preserved, different parametrisations of the same surface result in the same value of the vector surface integral.
- Vector surface integrals over surfaces that consist of several non-overlapping simple surfaces can be evaluated as the sum of vector surface integrals over the pieces, given that the orientation of the normals is consistent.